Valentin François



Final-year engineering internship

Mechanical design & product development

Availability · Autumn 2027

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Selected projects

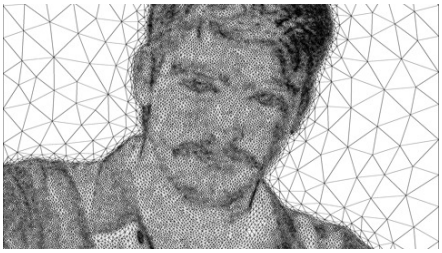
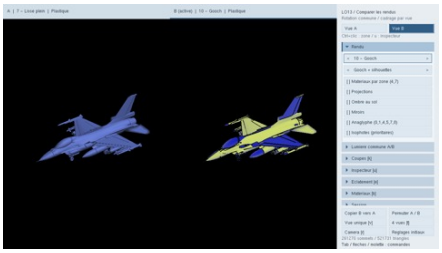


Image-based meshing

Python · Gmsh

Image-based triangular mesh generation with refinement driven by image variations.



3D graphics

C · OpenGL

3D geometry visualization: cross-sections and surface analysis using isophotes.



Radio-controlled boat

CAD · CNC · 3D printing

Design and fabrication from plans. Motorized crane with camera feedback.

Experience

Engineering Data · Engineering internship

Fondettes · 2025

Design and sizing of mechanical systems for robotic production lines. Analytical calculations and finite element analysis (SolidWorks Simulation). 3D-printed prototyping, assembly and in-house testing followed by on-site customer trials.

COSTIER STC · Shop-floor internship

Perpignan · 2023

Updated technical documentation and operating and maintenance manuals.

Model building

Association Modélisme Stéphanois · 2010 - present

Design and fabrication of scale models and animated layouts. CAD, CNC milling, laser cutting, 3D printing. Construction of controlled and automated mechanical systems. Delivery of training sessions for association members.

Education

Mechanical engineering · 2022–2027

French engineering degree · UTT

Exchange semester · 2027

Cassino · Italy

Mechanics and in-service performance of materials and products · 2026

Master's degree · UTT

Exchange in Mechanical Engineering Technology · 2024

Cégep de Sherbrooke · Canada

Aeronautical initiation certificate (BIA) · 2020

Awarded with merit

Skills

CAD: Creo, SolidWorks · Numerical simulation: Abaqus, SolidWorks Simulation
Mechanical sizing · Python, C · Turning, milling · Prototyping

English B2+ · Spanish B1 · Driving licence B

Portfolio

valentin-francois.github.io